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22. No. of moles of carbon present in 6 g of C-12 = Mass / Molar mass

= 6/12 = 0.5 moles = 0.5 × 6.022 × 10^{23} carbon atoms.

No. of moles in 8 g of methane = 8 / 16 = 0.5 moles

= 0.5 × 6.022 × 10^{23} carbon atoms.

No. of moles in 7.5 g of ethane = 7.5 / 16 = 0.25 moles

= 2 × 0.25 × 6.022 × 10^{23} carbon atoms.

23. Percentage of carbon in ethylene $(C_2H_4) = \frac{\text{mass of carbon}}{\text{Molar mass}} \times 100$

$$= \frac{24}{28} \times 100 = 85.71\,\%$$

Percentage of carbon in propene $(C_3H_6) = \frac{36}{42} \times 100 = 85.71\%$

24. (a) relative atomic mass of C-12 is 12 u

25. $_6C^{12}$

II Key to brief answer questions:

32. Given :

The density of CO_2 at 273 K and 1 atm pressure = 1.965 kgm^{-3}

Molar mass of CO_2 = ?

At 273 K and 1 atm pressure, 1 mole of CO_2 occupies a volume of 22.4 L

Mass of 1 mole of $CO_2 = \frac{1.965\ \text{Kg}}{1\text{m}^3} \times 22.4\ \text{L}$

$$= \frac{1.965 \times 10^3\ \text{g} \times 22.4 \times 10^{-3}\ \cancel{\text{m}^3}}{1\ \cancel{\text{m}^3}}$$

$$= 44.01\ \text{g}$$

Molar mass of CO_2 = 44 $gmol^{-1}$